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Free-living monitoring of ALS progression in upper limbs using wearable accelerometers

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Abstract

Background Wearable technology offers objective and remote quantification of disease progression in neurological diseases such as amyotrophic lateral sclerosis (ALS). Large population studies are needed to determine generalization and reproducibility of findings from pilot studies.

Methods A large cohort of patients with ALS (N = 202) wore wearable accelerometers on their dominant and non-dominant wrists for a week every two to four weeks and self-entered the ALS Functional Rating Scale-Revised (ALSFRS-RSE) in similar time intervals. Wearable device data were processed to quantify digital biomarkers on four upper limb movements: flexion, extension, supination, and pronation using previously developed and validated open-source methodology. In this study, we determined the association between digital biomarkers and disease progression, studied the impact of study design in terms of required sensor wear-time and sensor position, and determined the impact of self-reported disease onset location on upper limb movements.

Results The main investigation considered data from a sensor placed on the non-dominant wrist. Participants with higher ALSFRS-RSE scores performed more frequent and faster upper limb movements compared to participants with more advanced disease status. Digital biomarkers exhibited statistically significant change over time while their rate of change was more profound compared to survey responses. Using data from the dominant wrist and changing data inclusion criteria did not alter our findings. ALS disease onset location significantly impacted use of upper limbs. Results presented here were comparable to an earlier study on twenty patients with ALS.

Discussion Digital health technologies provide sensitive and objective means to quantify ALS disease progression. Interpretable approaches, such as the one used in this paper, can improve patient evaluation and hasten therapeutic development.

Introduction

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease that causes progressive weakness [1, 2]. The disease typically begins in one body region and spreads to adjacent body regions, though disseminated onset is not unheard of. Thus, the onset and pattern of spread can vary. In ALS trials, clinical outcome measures are used to monitor and quantify disease progression. The ALS Functional Rating Scale-Revised (ALSFRS-R) is the most commonly used clinical outcome measure in ALS trials, and while it performs well, it is subjective and ordinal [3–5]. The self-entry form of the ALSFRS-R (ALSFRS-RSE)

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is commonly used in disseminated ALS research and correlates closely with the ALSFRS-R [5–12].

Digital clinical outcome assessments (COAs) use passive data collection from wearable devices and statistical and machine learning algorithms to analyze continuously collected data to produce objective measures of the extent of limb movement and its characteristics [13–15]. These wearable devices are used in ALS research to quantify behavioral and functional change over time as the disease progresses. Previously quantified features include total activity volume, vertical movement index, active versus sedentary time, time spent at home, step count, and gait cadence [7, 11, 16–21].

Recently, we demonstrated the utility of digital COAs for ALS monitoring to quantify four fundamental upper limb movements: flexion, extension, supination, and pronation. Using a single, wrist-worn, tri-axial accelerometer worn on the non-dominant limb in a cohort of 20 people living with ALS (PALS), we demonstrated that these digital COAs quantified changes in movement in the upper extremities [22].

In this study, we reproduce our earlier findings in a larger independent cohort of over 200 PALS using the same methodology, extend our investigation to movements of the dominant limb, relax the daily sensor wear-time compliance, and compare the ratio between dominant and non-dominant limb metrics versus self-reported disease onset.

Methods

Ethical considerations

The study was conducted in accordance with the ethical principles posited in the Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Subjects. The protocol was approved by the institutional review board (Advarra Center for IRB Intelligence (CIRBI)). Participants underwent the consent process and provided documentation of informed consent prior to any study procedures. There was no participant compensation.

Participants and data

The data was collected by ALS Therapy Development Institute (ALSTDI) as a part of the ALS Research Collaborative (ARC) Study which has enrolled over 600 PALS and assembled a rich dataset including self-reported ALSFRS-R (ALSFRS-RSE) scores together with digital physiologic data, speech recordings and accelerometer measurements, and an accompanying biorepository including skin biopsies, whole genome sequences, and plasma samples [17].

For this analysis, we included 438 PALS who participated between September 2014 and January 2023 and

simultaneously wore two accelerometers on non-dominant and dominant wrists, specifically the ActiGraph GT3X+ (ActiGraph, Pensacola, FL), for one week every two to four weeks. According to the study protocol, wearing the device during nighttime was optional and participants could take it off for any reason, including discomfort.

The devices collected continuous triaxial accelerometer measurements with sampling frequency of 30 Hz and dynamic range of ± 6 g (gravitational units). Handedness information and disease onset (dominant, non-dominant upper limb) was self-reported and collected using a web-based survey presented to participants through their password protected ARC Study web portal. ALSFRS-RSE scores were collected using a web-based platform that presented the survey to them, as previously described [17].

Capturing upper limb movements using accelerometer data

We used raw sub-second level accelerometer data from both wrists to quantify upper limb movement metrics with our previously described method [22]. In short, the approximated gravitational component of the raw accelerometer data is used to detect monotonic changes in sensor orientation that are characteristic of distinct upper limb movements. In this way, specific movements can be detected and their number and duration quantified. Specifically, we analyzed count and duration metrics for six movements: 1) upper limb flexion, 2) extension, 3) supination, 4) pronation, 5) combined flexion and extension, and 6) combined supination and pronation exceeding a threshold of 45 degrees. Count and duration metrics were also computed for these movements using angular thresholds of 90 and 135 degrees. These three thresholds, 45, 90, and 135 degrees, were intended to capture movements that are narrow, moderate, and extensive in range, respectively. Based on our prior work, the 45-degree threshold captures most movements accurately, and we used this threshold in our primary analysis; 90- and 135-degree thresholds were used in our sensitivity analyses (available in Sect. "Methods" in Supplementary Materials). Statistics for each type of movement were aggregated daily. Duration metrics were quantified from 10 fastest movements of a given type and expressed in seconds. We report the cumulative daily count and average duration for each movement over the day.

Upper limb movements are denoted as X_{yz} , where X represents a type of metric: C stands for cumulative daily count and D for average duration across the day. The first subscript y represents a type of movement: f for flexion, e for extension, s for supination, and p for pronation; the second subscript z represents the threshold angle

in degrees. For example, C_{f45} is the cumulatively daily count of limb flexions by at least 45 degrees. In addition, we defined metrics that combine flexion and extension, as well as supination and pronation. These metrics use a dual notation for y , e.g., C_{fe45} refers to the total count of limb flexions and extensions by at least 45 degrees. Full list of upper limb movement notations is provided in Table S1 (Supplementary Materials).

Statistical analysis

The analytic sample included wearable device data and survey scores from participants with at least two complete ALSFRS-RSE surveys. The number of complete ALSFRS-RSE survey submissions and average time between the surveys were computed for each participant, and the overall survey scores were characterized using mean, standard deviation (SD), median, and range.

Sensor wear-time was determined using a previously described algorithm [23]. To ensure a more uniform wear-time across participants over their monitoring period, a participant day was deemed valid if there was at least 21 h (1260 min) of accumulated wear-time that day. Sensor wear compliance was calculated as the number of valid days divided by the number of days with accelerometer data collection. The follow-up duration was calculated as the time elapsed between the first and last day of accelerometer data collection.

The ALSFRS-RSE is a 12-item survey with each item scored 0–4 points [24]. Higher scores denote better function. We defined the following outcomes based on the ALSFRS-RSE total and disjoint subdomain scores: total score (Q1–12), bulbar subdomain (Q1–3), fine motor subdomain (Q4–6), gross motor subdomain (Q7–9), and respiratory subdomain (Q10–12).

We computed participants' average daily accelerometer-based metrics over the 7 days before and 7 days after the baseline survey (the first ALSFRS-RSE survey). We calculated correlation (Pearson r) for each pair of baseline metrics: 12 upper limb movement metrics (using the 45-degree threshold), Q1–12 (total score), Q1–3 (bulbar subdomain), Q4–6 (fine motor subdomain), Q7–9 (gross motor subdomain), and Q10–12 (respiratory subdomain).

We fitted linear mixed-effect models (LMMs) to estimate baseline and monthly rates of change of ALSFRS-RSE scores and accelerometer-based upper limb movement metrics. In each model, we specified time (months elapsed from the start) as a fixed effect and included participant-specific random intercept and random slope.

The LMMs were also used to assess the association between the upper limb movement metrics and ALSFRS-RSE scores. The five outcomes (one per model) were the

full and partial ALSFRS-RSE scores (total score; bulbar, fine motor, gross motor, and respiratory subdomains). The sole covariate was a fixed-effect metric of the average daily movement values from 7 days before and 7 days after the time-matched survey.

The main investigation was carried out for each of the 12 upper limb movement metrics using a 45-degree threshold. This resulted in a total of 60 models (5 survey scores $\times$ 12 upper limb movement metrics per threshold) being fit. Sensitivity analyses were evaluated for movements defined by 90- and 135-degree thresholds and wear-time thresholds of 16, 8, and 4 h (Sect. 2.3 in Supplementary Materials). The generalizability of the approach to dominant wrist data is also presented in Sect. "Results" in Supplementary Materials.

Finally, we computed the ratio between dominant and non-dominant limb metrics, denoted with " r " preceding the metric notation: e.g., metric rC_{f45} indicates a ratio between counts of dominant and non-dominant forearm flexions by at least 45 degrees. The ratio was calculated for each valid day of simultaneous wear of two sensors and averaged over the first month of observation. The natural logarithm was used to ease interpretation of the ratios, i.e., values of the count metrics that were positive indicated more frequent movements of the dominant limb, while negative values of the duration metrics indicated faster moves of the non-dominant limb. Two sample t -tests were used to compare ratio metrics derived from three groups of participants, i.e., those with ALS onset reported on dominant upper limb, non-dominant upper limb, or neither. When appropriate, we report 95% confidence intervals (CIs) for fixed effects estimates. In our analysis, we accounted for multiple testing using the conservative Bonferroni correction for the three threshold angles; starting with a nominal Type I error rate of $\alpha = 0.05$ and with $m = 36$ tests (12 per threshold), the resulting adjusted cutoff p -value for statistical significance is 0.0003 [25].

Results

Demographics, compliance, ALSFRS-RSE status

Out of 438 participants (Table 1) with any accelerometer data collected, the information on handedness was available for 354 participants (Fig. 1), while ALSFRS-RSE surveys were available for 310 participants. A total of 108 participants had insufficient sensor wear-time (less than 21 h per day) and these participants were excluded from the subsequent analyses. After accounting for valid accelerometer data, the analytic dataset consisted of data for 202 participants with a mean (SD) follow-up of 895.0 (694.9) days, a mean (SD) number of collected accelerometer data days of 282.8 (215.5), a mean (SD) number of valid accelerometer data days of 51.34 (64.73), a number

Table 1 Baseline demographics and descriptive statistics for ALSFRS-RSE and wearable device data in the analytic sample

	All participants	Participants in analytic dataset
Population characteristics		
Total number of participants, N	438	202
Demographics		
Age, mean (SD), median [min, max]	54.46 (10.71), 56 [20, 78]	54.84 (10.9), 57 [20, 78]
Males, N (%)	281 (64.16)	127 (62.87)
Female, N (%)	157 (35.84)	75 (37.13)
Race		
White, N (%)	408 (93.15)	189 (94.03)
Hispanic, N (%)	10 (2.23)	6 (2.99)
Asian, N (%)	8 (1.83)	4 (1.99)
Black, N (%)	3 (0.68)	2 (0.99)
Other, N (%)	6 (1.37)	0 (0)
Handedness		
Right, N (%)	306 (69.86)	173 (85.64)
Left, N (%)	47 (10.73)	28 (13.86)
Either, N (%)	1 (0.23)	1 (0.50)
Unknown, N (%)	84 (19.17)	-
Clinical characteristics		
Baseline ALSFRS-RSE Q1—12, mean (SD), points	40.75 (5.31)	40.83 (5.15)
Delta ALSFRS-RSE, mean (SD), points per month	-0.714 (0.816)	-0.570 (0.541)
Diagnosis delay*, mean (SD), months	16.27 (20.93)	17.50 (20.78)
Symptom duration at sign up (SD), months	28.24 (30.51)	33.61 (36.09)
ALS onset		
Non-dominant upper limb, N (%)	-	32 (15.84)
Dominant upper limb, N (%)	-	29 (14.36)
Both upper limbs, N (%)	-	16 (7.92)
Neither upper limb, N (%)	-	125 (61.88)
Outcome and accelerometer data		
ALSFRS-RSE		
Number of surveys, mean (SD), count	12.65 (12.34)	15.98 (14.65)
Time between surveys, mean (SD), days	65.48 (35.05)	65.97 (34.09)
Accelerometer data		
Follow-up, mean (SD), days	776.0 (654.7)	895.0 (694.9)
Collected accelerometer data, mean (SD), days	248.7 (192.9)	282.8 (215.5)
Valid accelerometer data, non-dominant wrist, mean (SD), days	-	48.40 (60.87)
Valid accelerometer data, dominant wrist, mean (SD), days	-	51.34 (64.73)

For compliance, participant-specific values are aggregated across all study participants. A valid day in the observation period was defined as a day with at least 1260 min (21 h) of wear time

*Calculated as time elapsed between symptom onset and formal diagnosis

of ALSFRS-RSE surveys of 15.98 (14.65), and a mean time between surveys of 65.97 (34.09) days.

Upper limb movements metrics

Baseline correlations

Baseline movement counts had a moderate positive correlation with the baseline ALSFRS-RSE total scores (Pearson $r \geq 0.43$), whereas movement duration had a

moderate negative correlation ($r \leq -0.48$) (Fig. 2) [26]. PALS with higher overall levels of functioning on the ALSFRS-RSE performed more movements, and the movements were faster compared to individuals with more advanced disease. Movement metrics were moderately correlated with survey subdomains for gross (Q4—6) and fine motor functions (Q7—9) but negligible for bulbar (Q1—3) and respiratory (Q10—12) functions.

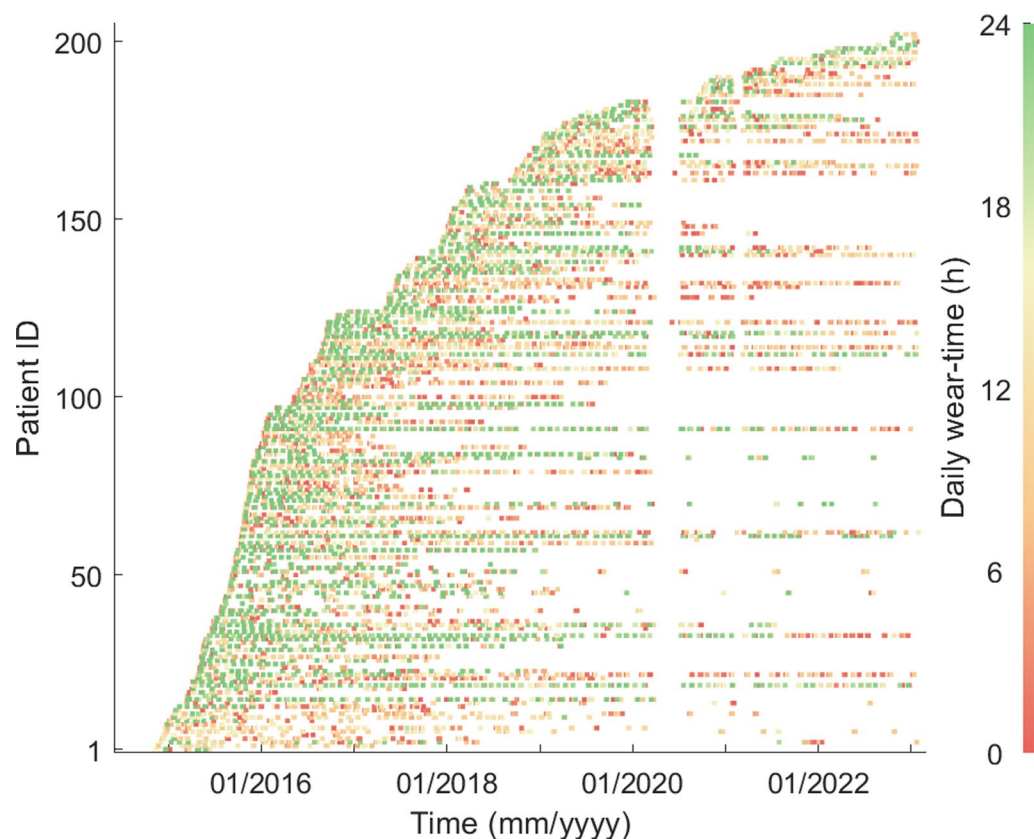


Fig. 1 Participant enrollment and data collection from the non-dominant wrist. Figure presents participants included in analytic dataset only. Dot color indicates daily sensor wear-time. To improve readability, we removed points with 0 min of wear-time

Baseline and monthly change

Linear mixed effects models (LMMs) were used to quantify average baseline values and monthly change of the daily metrics of limb movements (Table 2). At baseline, counts for daily flexion (721.5) and extension (721.7) were lower than supination (1640) and pronation (1694). Also at baseline, limb flexions and extensions took longer (0.545 s and 0.543 s) compared to supinations and pronations (0.347 s and 0.376 s, respectively). Over time, daily count metrics showed a significant decrease while duration of movements increased significantly (Table 2). Notably, the relative rate of change was greater in all movement measures compared to ALSFRS-RSE scores, however the signal-to-noise ratio indicated similar strength only for the count measures. Results for the alternative angular thresholds are provided in Table S2 (Supplementary Materials).

Association with ALSFRS-RSE Q1–12 and subdomains

All daily upper limb movement measures were significantly associated with total ALSFRS-RSE scores (Table 3). The count metrics decreased and duration metrics increased over time as the disease progressed.

Significant associations were also observed between upper limb movement measures and the ALSFRS-RSE subdomains. P-values were lower for fine motor and gross motor subdomains compared to the bulbar and respiratory subdomains, as expected (Fig. 3). The results on the association between upper limb movements metrics with the alternative angular thresholds and Q1–12 are provided in Table S3 (Supplementary Materials). The results for subdomains are available in Table S4 (Supplementary Materials).

Impact of disease onset

Data from the first month of observation were used to compare individuals with different disease onset in upper limbs. Valid accelerometer days were available for 32 participants with onset reported on non-dominant limb, 29 participants with onset reported on dominant limb, and 125 participants with disease onset that did not affect the upper limbs.

Statistically significant differences between groups with non-dominant and dominant limb onset were observed for all count and duration metrics with angle threshold of 45 degrees (see Table S9 in Supplementary Materials).

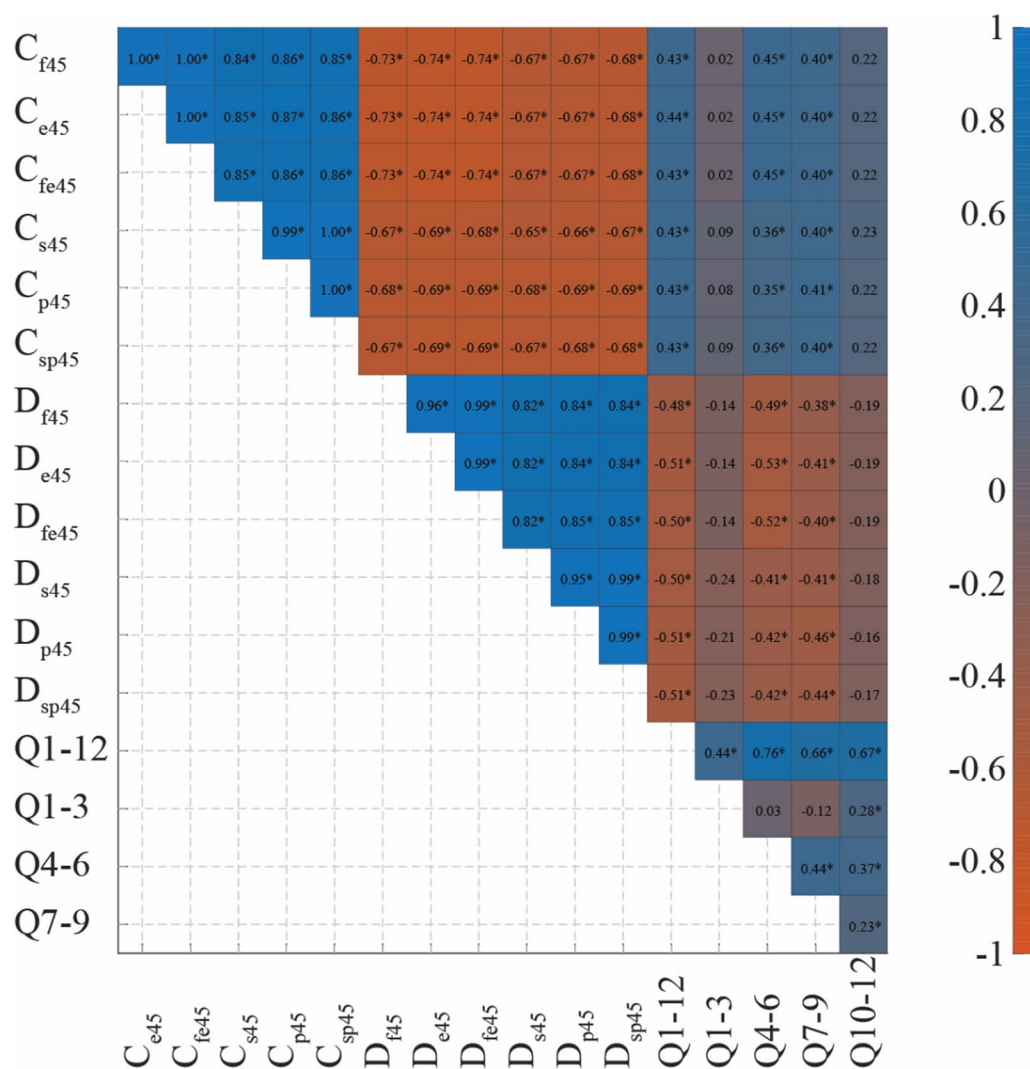


Fig. 2 Pearson correlation coefficient between participant-specific baseline values (for angle threshold 45 degrees). For each daily metric, its baseline value was calculated as a participant's average values recorded ± 7 days from the baseline ALSFRS-RSE. Note: 'C' indicates count metrics, 'D' duration metrics, 'Q' responses to questions in ALSFRS-R, 'f' upper limb flexions, 'e' upper limb extensions, 's' upper limb supinations, 'p' upper limb pronations, '45' angular threshold and '*' statistical significance

All metrics differentiate among participants with disease onset on dominant limb and those with no onset on upper limbs, while on the count metrics differentiate among participants with onset affecting non-dominant limb or neither (Fig. 4).

Moreover, the comparison between upper limb movement metric ratios in PALS with different disease onset reveals three results: (1) in PALS with disease onset in the non-dominant upper limb, the movements of the dominant upper limb are faster and more frequent; (2) in those with ALS onset in the dominant upper limb, the movements of the faster and more frequent in the non-dominant limb; (3) in those without onset on either, the ratio between use of dominant and non-dominant

is close to zero. For example, the log change in rC_{fe45} (ratio of upper limb flexions and extensions by at least 45 degrees) was 0.41 [0.23, 0.60] for participants with onset on the non-dominant limb, -0.27 [-0.40 , -0.13] for participants with onset on the dominant limb was, and 0.02 [-0.03 , 0.07] for without onset on either limb.

Discussion

Recent advances in sensing technologies for monitoring medical conditions have ushered in a new era for health research and healthcare, enabling clinicians to remotely monitor patients from a distance. This has reduced patient burden and enhanced the objectivity of measurements. Actigraphy from wearable devices have primarily

Table 2 Average baseline and monthly change in ALSFRS-RSE scores and the upper limb movement daily metrics

Outcome	Baseline estimate [95% CI]	Monthly change estimate [95% CI] (p-val.)	Relative change	Signal-to-noise ratio
ALSFRS-RSE scores				
Q1—12	41.54 [40.89, 42.19]	− 0.637 [− 0.714, − 0.562] (<0.0001)	− 1.62	− 1.03
Q1—3	10.78 [10.54, 11.01]	− 0.125 [− 0.147, − 0.102] (<0.0001)	− 1.23	− 0.68
Q4—6	9.83 [9.55, 10.11]	− 0.196 [− 0.220, − 0.172] (<0.0001)	− 2.10	− 1.04
Q7—9	9.44 [9.15, 9.73]	− 0.207 [− 0.231, − 0.183] (<0.0001)	− 2.32	− 1.10
Q10—12	11.34 [11.20, 11.56]	− 0.087 [− 0.105, − 0.069] (<0.0001)	− 0.80	− 0.61
Movement biomarkers				
C_{f45}	721.5 [676, 767.1]	− 19.55 [− 22.78, − 16.31] (<0.0001)	− 2.71	− 1.08
C_{e45}	721.7 [676.4, 766.9]	− 19.78 [− 23.04, − 16.51] (<0.0001)	− 2.74	− 1.08
C_{fe45}	1443 [1353, 1534]	− 39.36 [− 45.87, − 32.86] (<0.0001)	− 2.72	− 1.08
C_{s45}	1640 [1538, 1741]	− 41.94 [− 49.78, − 34.10] (<0.0001)	− 2.56	− 0.95
C_{p45}	1694 [1592, 1796]	− 43.35 [− 51.34, − 35.37] (<0.0001)	− 2.56	− 0.97
C_{sp45}	3334 [3131, 3537]	− 85.40 [− 101.24, − 69.57] (<0.0001)	− 2.56	− 0.96
D_{f45}	0.545 [0.520, 0.569]	0.016 [0.011, 0.021] (<0.0001)	2.88	0.53
D_{e45}	0.543 [0.519, 0.567]	0.017 [0.012, 0.022] (<0.0001)	3.09	0.53
D_{fe45}	0.543 [0.519, 0.567]	0.016 [0.011, 0.022] (<0.0001)	3.02	0.52
D_{s45}	0.347 [0.332, 0.363]	0.012 [0.009, 0.015] (<0.0001)	3.37	0.60
D_{p45}	0.376 [0.357, 0.394]	0.012 [0.008, 0.015] (<0.0001)	3.11	0.56
D_{sp45}	0.361 [0.344, 0.379]	0.012 [0.009, 0.016] (<0.0001)	3.42	0.58

Relative change was calculated as a ratio between monthly change estimate and baseline estimate expressed in percent. Signal-to-noise was calculated as ratio between the mean progression rate (fixed effect slope) and standard deviation of random slopes

'CI'—confidence interval; 'p-val'—p-value associated with a coefficient estimate; 'C'—count metrics; 'D'—duration metrics; 'Q' responses to questions in ALSFRS-R; 'f' upper limb flexions; 'e' upper limb extensions; 's' upper limb supinations; 'p' upper limb pronations; '45' angular threshold

Table 3 Model intercept and estimated average outcome change associated with a one-point increase in ALSFRS-RSE total score (Q1—12)

Movement biomarker	Intercept [95% CI]	Slope [95% CI] (p-val.)
C_{f45}	30.92 [29.33, 32.51]	0.012 [0.010, 0.014] (<0.0001)
C_{e45}	30.91 [29.32, 32.50]	0.012 [0.010, 0.014] (<0.0001)
C_{fe45}	30.89 [29.3, 32.48]	0.006 [0.005, 0.007] (<0.0001)
C_{s45}	30.98 [29.3, 32.66]	0.005 [0.004, 0.006] (<0.0001)
C_{p45}	30.68 [28.95, 32.41]	0.005 [0.004, 0.006] (<0.0001)
C_{sp45}	30.79 [29.08, 32.49]	0.002 [0.002, 0.003] (<0.0001)
D_{f45}	46.23 [44.34, 48.11]	− 14.850 [− 17.916, − 11.784] (<0.0001)
D_{e45}	46.31 [44.45, 48.17]	− 14.937 [− 17.992, − 11.883] (<0.0001)
D_{fe45}	47.03 [45.05, 49.00]	− 16.151 [− 19.412, − 12.890] (<0.0001)
D_{s45}	45.37 [43.59, 47.14]	− 21.361 [− 26.084, − 16.638] (<0.0001)
D_{p45}	45.6 [43.61, 47.58]	− 20.665 [− 25.774, − 15.555] (<0.0001)
D_{sp45}	46.22 [44.23, 48.2]	− 22.881 [− 28.130, − 17.633] (<0.0001)

'CI'—confidence interval; 'p-val'—p-value associated with a coefficient estimate; 'C'—count metrics; 'D'—duration metrics; 'Q' responses to questions in ALSFRS-R; 'f' upper limb flexions; 'e' upper limb extensions; 's' upper limb supinations; 'p' upper limb pronations; '45' angular threshold

focused on overall activity (time spent active, sedentary behavior, or gait), often neglecting individual limb function due to the complexity of analyzing limb movements in real-life and the heterogeneity of data.

In this study of data collected through ALS TDI's ARC Study, we applied our recent method for quantifying upper limb movements using wrist-worn devices in ALS [22]. As we found in our prior study, these upper limb movements effectively quantified functional decline over time, demonstrating statistically significant declines. Furthermore, changes in upper limb movements once again correlated with declines in the ALSFRS-RSE total score (Q1—12) and subdomains for fine motor (Q4—6), gross motor (Q7—9), and, to a lesser degree, bulbar (Q1—3) and respiratory (Q10—12) functions. Sensitivity analyses from movement metrics with angle thresholds of 90 and 135 degrees supported our primary analysis of movements defined by a 45-degree threshold.

These analyses represent an important validation of our prior work using wearable accelerometers to quantify upper limb movements in an independent cohort in people living with ALS. In this study, consistent with our previous findings, these measurements quantify a decline in movement that correlates with the ALSFRS-RSE, the gold standard used to assess ALS progression.

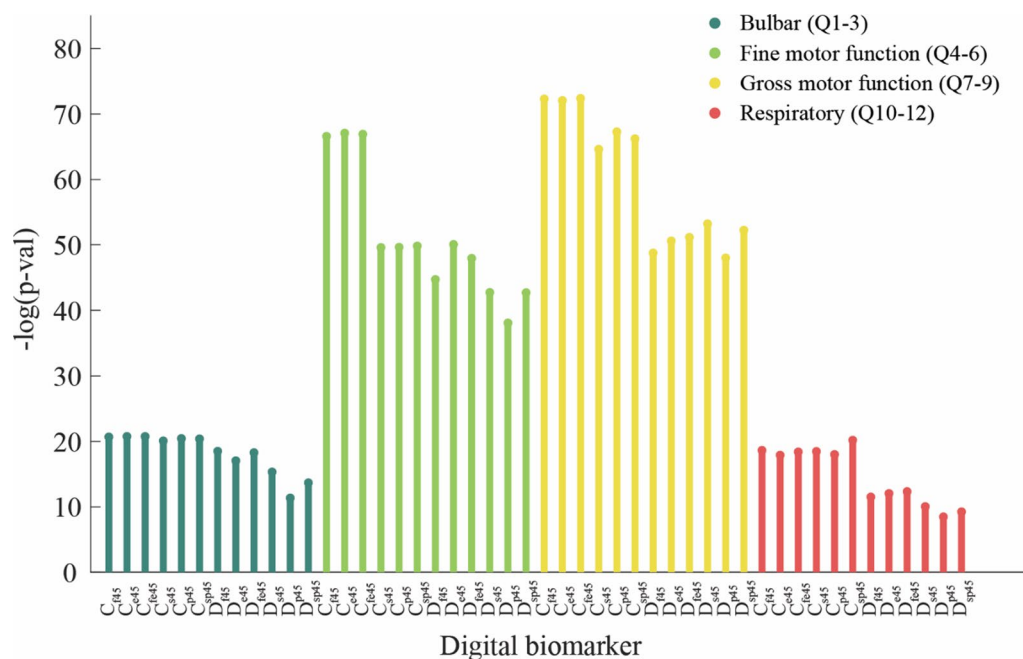


Fig. 3 Negative logarithm of p-values of upper body metrics' slopes in four subdomains of ALSFRS-RSE surveys: bulbar (Q1—3), fine motor (Q4—6), gross motor (Q7—9), and respiratory (Q10—12). Longer lines indicate lower p-value. Note: 'C' indicates count metrics, 'D' duration metrics, 'Q' responses to questions in ALSFRS-R, 'f' upper limb flexions, 'e' upper limb extensions, 's' upper limb supinations, 'p' upper limb pronations, and '45' angular threshold

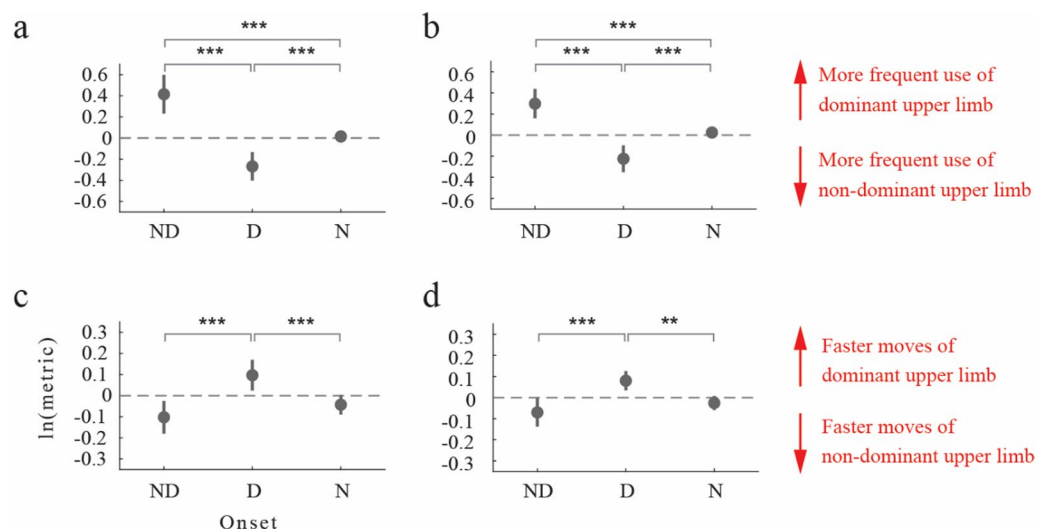


Fig. 4 Mean and 95% confidence intervals of four selected upper movement ratios between dominant and non-dominant upper limb movement metrics, rC_{fe45} (panel **a**), rC_{sp45} (panel **b**), rD_{fe45} (panel **c**), and rD_{sp45} (panel **d**), stratified by disease onset on non-dominant upper limb (ND), dominant upper limb (D), and neither (N). For count metrics (panels **a** and **b**), values greater than 0 indicate a more frequent use of the dominant upper limb; for duration metrics (panels **c** and **d**), values greater than 0 indicate faster movements of the non-dominant upper limb. The statistical significance of two sample t-test p-values are denoted as follows: * < 0.05, ** < 0.005, *** < 0.0005

The monthly decline of ALSFRS-RSE in this study and our prior study were similar, (-0.637 vs. -0.757 , respectively), despite different baseline ALSFRS-RSE estimates (41.54 vs. 34.37). Similar trends of functional decline

were also found in upper limb movement metrics. Specifically, the count metrics exhibited a significant decrease while the duration metrics a significant increase over time. This study had a much larger sample (202 vs. 20

in our prior study) and longer follow-up (29 months vs. 6 months), providing more statistical power to quantify changes. Here, the rates of change for arm movements were similar in direction, but smaller in magnitude for the count metrics (e.g., for C_{f45} : -2.71% vs. -4.86%) and greater for the duration metrics than reported previously (e.g., for D_{f45} : 2.88% vs. 1.70%). However, the two studies produced similar estimates of the association between daily measures and ALSFRS-RSE.

In our sensitivity analysis, we determined that a threshold on the required sensor wear-time did not play a major role in detecting trajectories of disease progression over time, but it did in estimating of average outcome change associated with ALSFRS-RSE responses (for details, see Sect. 2.3 in Supplementary Materials).

Our findings also generalize to the data collected from the dominant limb. Interestingly, baseline estimates of count and duration metrics of both upper limbs were similar. Count metrics from dominant limb were on average 3% greater than non-dominant, while duration metrics were on average 3% lower than the corresponding metrics from non-dominant limb. These results may indicate that the investigated approach does not distinguish more granular upper limb activities typically attributed to the dominant hand, such as handwriting. Alternatively, it may also suggest that this population does not use their dominant hand preferentially, as is seen in the general population [27].

In turn, significant differences in the frequency and duration of upper limb functioning were observed for individuals with ALS onset in one limb. Specifically, participants with onset in the non-dominant upper limb exhibited an average log change for the ratio of count and duration measures of 0.35 and -0.09 , respectively, indicating more frequent and faster movements of their dominant limb. Conversely, participants with onset in the dominant upper limb showed respective statistics of -0.25 and 0.09 , highlighting more frequent and faster movements of their non-dominant limb.

Our approach has several strengths. First, it uses data collected with a single accelerometer per wrist, unconstrained by initial location (left or right wrist), placement (top or bottom of the wrist), and orientation (anterior or posterior) [27], which makes it convenient for applications in free-living settings, even when technology adherence is limited. Second, it provides intuitive link to human kinesiology and aids in the conceptualization of disease progression on the limb of interest. Third, the investigated metrics of limb movements demonstrate significant (and greater than survey scores) decline over time as well as strong association with functional ALSFRS-RSE scores, highlighting their relevance in monitoring disease progression, especially of fine and gross

motor functions. Finally, using data from two sensors worn simultaneously allows identification of upper limb disease onset remotely and objectively.

Our study also has some limitations. First, not every day was quantified, with the average number of valid accelerometer days being only 48.4 compared to a mean data collection period of 282.8 days. This discrepancy may be likely due to several factors, including the study design, participants' compliance to device wear, and a conservative data inclusion criterion. In this study, the data collection occurred in irregular intervals, e.g., for a week every three weeks. Although disadvantageous to data completeness, this approach was required to secure time for data downloading, device battery recharging, and importantly minimizing participant burden and likelihood of drop-out. Participants could also take off the device for nighttime or at any given time of a day as needed, which resulted in a substantial non-wear time and limited data for analysis, especially given our use of valid day criterion of 21 h of device wear-time. Finally, our method is based on recordings of just one accelerometer and, therefore, it only allows for an approximation of limb movements unaffected by significant linear acceleration. An improved approach should consider fusing of data from additional sensors, in particular gyroscope and magnetometer (unavailable in ActiGraph GT3X+ used in this study).

This study serves as a step forward in the process toward clinical validation of upper limb movement metrics as an objective measure of functional decline [29]. Translating these outcomes into clinically meaningful terms remain crucial. One approach is to calculate the Minimal Detectable Change (MDC), which quantifies the smallest amount of change that can be confidently attributed to real changes rather than measurement error [30, 31]. Calculating MDC for movement metrics at shorter time intervals, e.g., within a week to ensure no true change has occurred, would offer clinicians a clearer picture of significant changes over time. On the other hand, translating the outcomes into Minimal Important Difference (MID) or Minimal Clinically Important Difference (MCID) remains challenging. Future studies should incorporate patient-reported outcomes or clinical judgments as anchors for MIC/MCID calculations. Although this study was not designed to capture these impressions, we believe that future trials should incorporate questions and assessments associating limb function metrics with perceived functional changes at frequencies that would allow for calculation of both the MDC and MID/MCID.

A more direct way to bridge the gap between movement metrics and every-day functioning is to relate these metrics to common activities. For example, the metrics capturing upper limb flexion and extension

could be related to activities of personal hygiene or dressing, while the metrics on supination and pronation could be related to handling utensils or turning a doorknob. These interpretations could help clinicians better understand the functional impact of movement declines. Further research is needed to determine clinically relevant thresholds for changes in these metrics.

Although digital health technologies (DHTs) have not yet been widely implemented in ALS clinical trials, a notable progress has been made. At least one ALS clinical trial has used smartphones to collect mobility and speech metrics [32], and digital speed recordings have recently become part of the Healey ALS Platform Trial [33]. A primary barrier to use these technologies more broadly may be resulting from the small sample sized in prior studies [11, 16, 22, 34]. In contrast, this study, with its larger cohort, demonstrates that upper limb movement metrics remain robust to sample size providing additional evidence for the inclusion of DHTs in future trials. Our research also suggests that in future trials it will be essential to set more transparent expectations for device wear-time and to provide regular contact and reminders for study participants, as it has led to the improved adherence in previous studies [35, 36]. Notably, comparison using a more liberal wear-time threshold of 16 or 8 h, resulted in including 240 or 308 subjects, respectively, instead of 202 included in our main evaluation. In conclusion, we demonstrated that wrist-worn accelerometers, worn unilaterally or bilaterally, may serve as useful tools for quantifying upper limb ALS disease progression. These measurements can be combined with quantitative assessments of speech, lower limb function, and respiration for a more comprehensive evaluation of disease progression in ALS trial settings as clinical outcome assessments.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12984-024-01514-7>.

Additional file 1.

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Author contributions

Concept and design – MS, KMB, NC, AP, FGV, JPO, JDB. Data collection – AP, FGV. Statistical analysis – MS, JPO. Interpretation of results – MS, KMB, NC, JDB. Manuscript preparation – MS. Critical review of manuscript – KMB, NC, AP, FGV, JDB, JPO. Study supervision – JDB, JPO. JDB and JPO contributed equally.

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Availability of data and materials

Data may be shared upon request and after review and approval by the owners of the data. Shared data will consider deidentified summary statistics used for analytical analysis. Related correspondence should be sent to fvieira@als.net.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the ethical principles posited in the Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Subjects. The protocol was approved by the institutional review board (Advarra Center for IRB Intelligence (CIRBI)). Participants underwent the consent process and provided documentation of informed consent prior to any study procedures. There was no participant compensation.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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